Angioedema is an acquired or hereditary condition that is characterized by intermittent submucosal and subcutaneous edema formation. It often involves the gastrointestinal tract, face, and extremities. There are two types of angioedema. One results from the release of bradykinin and does not involve symptoms of an allergic reaction. The other is caused by the release of mast cell mediators and causes allergic reaction symptoms such as bronchospasm, flushing, urticaria, and hypotension.

The most prevalent hereditary form of angioedema is the result of dysfunction or an autosomal dominant deficiency of C1 esterase inhibitor. This protease inhibitor controls contact activation, complement, and fibrinolytic pathways. With the absence of C1 esterase inhibitor, vasoactive mediators such as bradykinin are released that produce edema and increased vascular permeability.

Patients who are deficient in C1 esterase inhibitor have recurrent episodes of facial and/or laryngeal edema that can last from 24 to 72 hours. They can be triggered by trauma, infection, menses, stress, or estrogen-containing oral contraceptives and usually begin in the second decade of life. Patients presenting with abdominal attacks experience nausea, vomiting, excruciating pain, and/or diarrhea. Laryngeal attacks can be triggered by dental surgery.

Patients with lymphoproliferative disorders can acquire C1 esterase inhibitor deficiency. These patients have C1 inhibitor antibodies that lead to a condition that resembles hereditary angioedema. ACE inhibitors used for the treatment of heart failure and hypertension may also precipitate angioedema in about 0.5% of patients. The increased availability of bradykinin brought about by the ACE inhibitor-mediated blockade of bradykinin catabolism appears to be the cause of this drug-induced angioedema.

Prophylactic therapy (both long-term and preoperatively) for patients with angioedema is achieved through the use of androgens such as stanozolol and danazol. It is believed that these drugs increase hepatic synthesis of C1 esterase inhibitor. Treatment with antifibrinolytic drugs has also been used. These drugs include: tranexamic acid, aprotinin, and aminocaproic acid, and exert their action by inhibiting the activation of plasmin.

An acute episode of angioedema is treated with 25 units/kg of C1 inhibitor concentrate or 2-4 units of FFP in order to replace the deficient enzyme. Acute episodes of angioedema are not effectively treated with androgens, antihistamines, catecholamines, and antifibrinolytics. Acute attacks leading to upper airway obstruction may necessitate tracheal intubation until the edema subsides. Adequate personnel and equipment should be available to perform a tracheostomy if needed when laryngoscopy is performed; however, tracheostomies in patients with massive airway edema may be impossible or very difficult to achieve.

Prophylactic treatment for patients with recurrent angioedema is necessary before any treatment requiring intubation or a stimulating procedure like dental surgery. Intravenous C1 inhibitor concentrates should be readily available to treat acute attacks. IM injections and regional anesthesia is tolerated well in this patient population.

Hines RL, Marschall KE. Stoeltings Anesthesia & Co-Existing Diseases. 6th ed. New York, NY: Churchill-Livingston; 2012: 521-522.

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